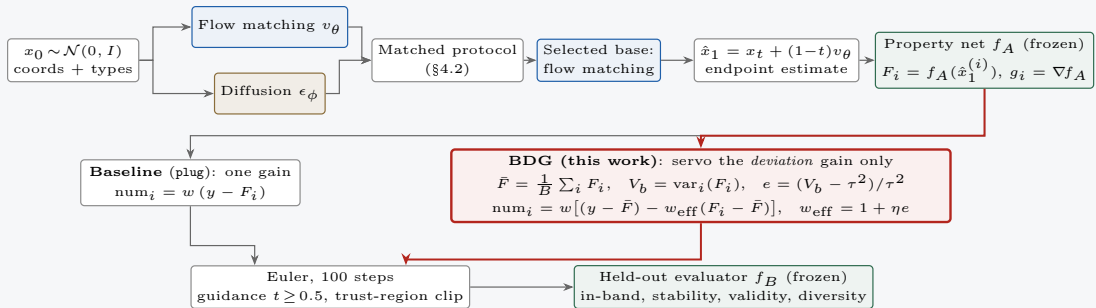
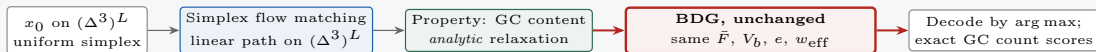


(a) Modality 1: QM9, 3D molecular coordinates $\mathbb{R}^{N \times 3}$



(b) Modality 2: DNA enhancers on the probability simplex $(\Delta^3)^L$



Reused unchanged: the controller, its state $(\bar{F}, V_b, e)$, the setpoint τ , the gain η , and the w_{eff} reduction.

Modality-specific: state space, generator, property map, decoder. The step is a per-sample scalar times plug's step, so no covariance matrix and no inverse are needed and the update stays tangent to the simplex.